Unit XV: Introduction to Biochemistry

Lecture notes: the slides plus what she explained in class (transcript) and wrote on the board

CHEMISTRY (PROF. L. RANAY)

WHAT BIOCHEMISTRY IS

- **Biochemistry** = the study of the **chemical substances found in living organisms** (plants, animals, humans) and the **chemical interactions** of these substances with each other.
- It covers the compounds, reactions and molecular interactions involved in the **production, maintenance and reproduction** of living organisms: how cells make the molecules needed for life and how the reactions that keep life going happen.
- All living things use the **same types of biomolecules** and use energy, so life can be studied with the methods of chemistry and physics.
- Cells and biomolecules are thought to have arisen from very simple molecules: **H_2O, CH_4, NH_3, N_2 and H_2**.
- Biochemistry draws on many disciplines and answers questions on the **molecular nature of life** (growth, reproduction, metabolism).

TWO KINDS OF CHEMICAL SUBSTANCES IN THE BODY

- **Bioinorganic substances: water** (about **70%** of human body mass) and **inorganic salts**.
- **Bioorganic substances** (contain **carbon**): **proteins, nucleic acids, lipids and carbohydrates**. These four are the **biomolecules**.
- Together, inside the cell, they undergo chemical interactions that **sustain life**.

Slide exercise

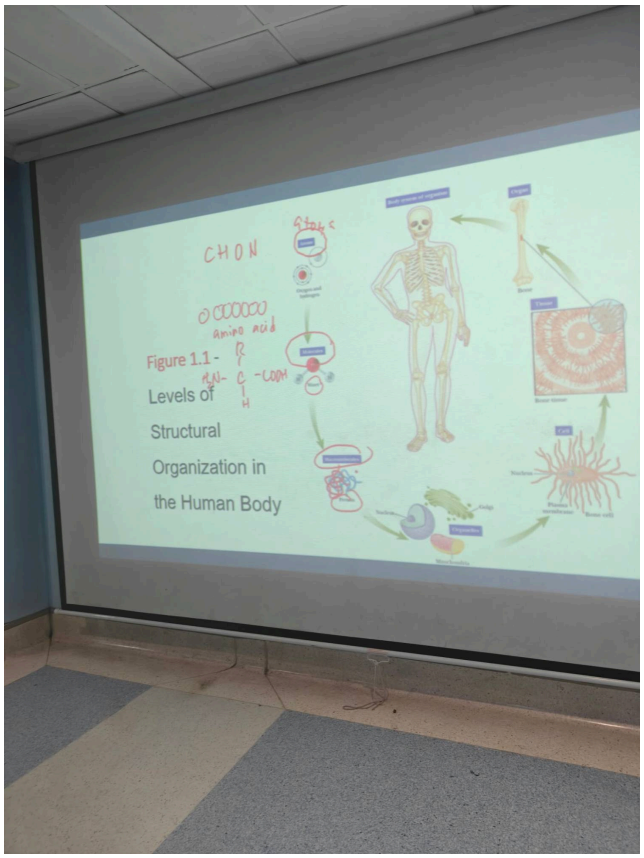
Biochemistry includes the study of **carbohydrates, lipids, proteins and nucleic acids** (answer c). The wrong choices drop nucleic acids or add salts.

BIOCHEMISTRY AND MEDICINE (HER EXAMPLES)

- **Nucleic acids** (DNA, RNA): storage and transmission of genetic information → **genetic diseases**.
- **Proteins:** the sequence (arrangement) of the structure → **sickle cell anemia**.
- **Lipids:** good and bad cholesterol. In the **lipid profile**, the bad cholesterol is **LDL (low-density lipoprotein)**; high LDL → **cardiovascular disease, atherosclerosis**.
- **Carbohydrates** (sugar): high blood glucose → **hyperglycemia, diabetes mellitus**.

LEVELS OF STRUCTURAL ORGANIZATION (FIGURE 1.1)

- Arranged from **smallest/simplest to largest/most complex**, put in order so the body functions efficiently:
- **Atoms → molecules → macromolecules → organelles → cells → tissues → organs → body system / organism**.
- **Atoms:** the four most common elements in the body are **C, H, O, N** ("CHON").
- **Molecules:** atoms share electrons in **covalent bonds**. Examples: water (bioinorganic), an **amino acid** (bioorganic; she drew $H_2N–CH(R)–COOH$).
- **Macromolecules:** amino acids bonded together get longer → polymer / **polypeptide chain** = **protein**.
- **Organelles** have distinct tasks: **mitochondria** = powerhouse of the cell; **nucleus** = control center / storage of genetic material.
- **Cell** (e.g., bone cell) = the **basic unit of life**. Cells → **tissue** (bone tissue) → **organ** (bone) → **system** (skeletal system: bones, joints, cartilage; functions: **support/structure, protection of organs, movement**).



Her board work on Figure 1.1: CHON, the amino acid $H_2N–CH(R)–COOH$, atoms → molecules → macromolecules circled

CHEMICAL FOUNDATIONS OF BIOCHEMISTRY

- Biochemistry describes the **structure, organization and function** of living organisms in **molecular terms**. To understand life at the molecular level you need:
  - knowledge of the **chemical structures** of the biomolecules;
  - understanding of their **biological functions**;
  - understanding of **bioenergetics** = the study of **energy flow in cells** (how energy from food is acquired, transformed and used to maintain life).
- **Functions she listed:** lipids = **long-term energy storage** and **membranes**; nucleic acids = **store and transmit genetic information**; carbohydrates = **energy storage**; proteins = many: **transport, structure, defense, catalysis**.
- **Organic chemistry** = the study of carbon compounds. The cellular apparatus is made of carbon compounds, so biomolecules belong to organic chemistry and their reactions are described by organic chemistry methods. **Carbon is the backbone** of every biomolecule.
- **Functional group** = an atom or group of atoms that shows characteristic physical and chemical properties. Her examples:
  - carbohydrates (glucose): **hydroxyl (–OH)** groups
  - lipids (fatty acids): **carbonyl** (part of the carboxyl group)
  - proteins (amino acids): **amino ($–NH_2$)** and **carboxyl (–COOH)**
  - DNA: **phosphate** group attached to the sugar; **ATP (adenosine triphosphate)** has **three** phosphate groups.

MONOMERS AND POLYMERS

- **Polymer** (Greek *poly* + *meros*, many parts); **monomer** (*mono* + *meros*, single part) = the **building block**.

SUMMARY TABLE: BUILDING BLOCKS AND MACROMOLECULES

Building block (micromolecule)	Macromolecule
Amino acid	Protein
Sugar (monosaccharide, e.g., glucose)	Carbohydrate (polysaccharide)
Nucleotide	Nucleic acid (DNA and RNA)
(none)	Lipids

- Carbohydrates: glucose + glucose → **disaccharide** → **oligosaccharide** → thousands = **polysaccharide** (e.g., **starch**).
- **Lipids have no common structure**, many different structures and functions, and **do not form polymers**. In the proteins lecture she stressed: **lipids are NOT macromolecules**; only proteins, nucleic acids and carbohydrates are.
- **Proteins and nucleic acids** play the key role in life processes.

INFORMATIONAL MACROMOLECULES (FIGURE 1.4)

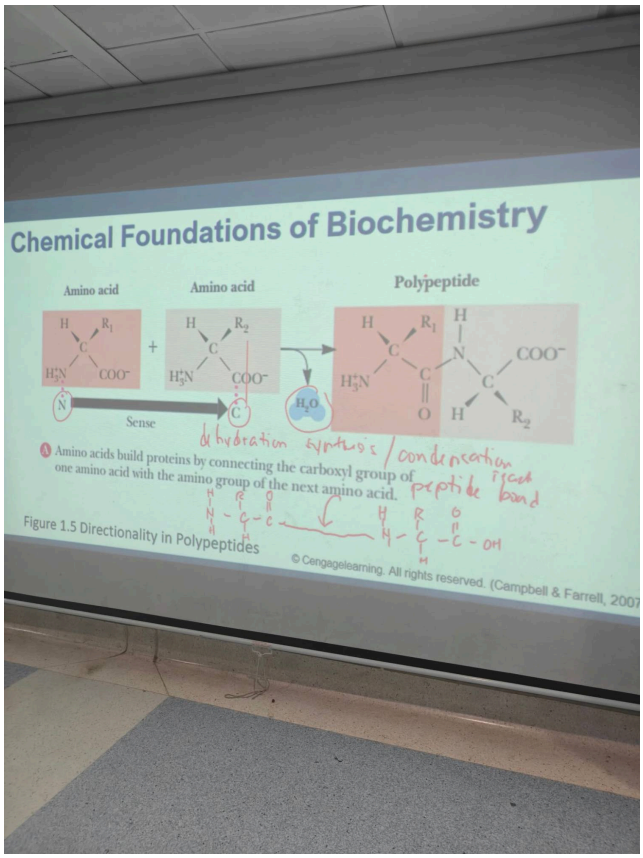
- In a DNA strand and a polypeptide, the sequence of building blocks is **not repeated**; in a polysaccharide the same building block (**Glc**, glucose) **repeats**.
- So **proteins and nucleic acids are the informational macromolecules**: they carry information. Central dogma: DNA is replicated, transcribed into RNA, and RNA is translated into protein.
- Polysaccharides are **not** informational (repeating glucose).

DIRECTIONALITY ("SENSE") OF THE BONDS (FIGURES 1.5 TO 1.7)

- Every bond that links building blocks has a **direction** (directionality, also called **sense**). In all three, the bond forms by **dehydration synthesis**, also called a **condensation reaction**: a **water molecule** is removed (an –OH from one unit and an –H from the other: H–OH = H_2O).

PROTEINS: PEPTIDE BOND

- Two amino acids have the same common structure (amino group, carboxyl group, H and R on the α-carbon); they differ only in the **R group**.
- The **carboxyl group** of one amino acid bonds with the **amino group** of the next: the **OH** leaves the carboxyl and an **H** leaves the amino group → **water**.
- What is left (the carbonyl carbon and the nitrogen) join by a covalent bond called the **peptide bond** (C–N).
- Direction: from the **N-terminal end** to the **C-terminal end** → polypeptide chain → protein.



Her board work on Figure 1.5: N → C (sense), H_2O removed, "dehydration synthesis / condensation", peptide bond

CARBOHYDRATES: GLYCOSIDIC BOND

- Glucose ring carbons are numbered C1 to C6. The bond links **carbon 1** of one glucose to **carbon 4** of the next (a **1→4** link); water is removed (dehydration synthesis).
- The bond is a **glycosidic bond**.

NUCLEIC ACIDS: PHOSPHODIESTER BOND

- A **nucleotide** has three parts: a **nitrogenous base**, a **pentose sugar**, and a **phosphate group** (ATP has three phosphates; a nucleotide of DNA/RNA has one).
- The **3' carbon** of the sugar carries an **–OH (hydroxyl)**; the **5' carbon** carries the **phosphate**. The next nucleotide links at the 3'–OH; read "prime": direction **5'** → **3'**.
- Water is removed; the bond is a **phosphodiester bond**.

> **Note:** Take note of the names of the bonds: **peptide** (proteins), **glycosidic** (carbohydrates), **phosphodiester** (nucleic acids). All three form by dehydration synthesis (condensation).

METABOLISM

- **Metabolism** = the **entire set of life-sustaining chemical reactions** in organisms; the **sum of all chemical processes** involved in maintaining life.
- Thousands of reactions: getting energy from food, processing and removing waste, building muscle, growth, photosynthesis in plants, cell division, reproduction.
- Over **40 years**, at nearly constant weight, the body processes about **6 tons of food** and **10,000 gallons of water**.
- The reactions are organized into smaller sets of **sequential** reactions (the product of one step is the reactant of the next) = **metabolic pathways**.
- The species produced along a pathway = **metabolites**.
- Many steps need **enzymes** (biological **catalysts** that speed up reactions), so organisms control pathways by **upregulating, downregulating, inhibiting or activating** enzymes.

ANABOLISM VS CATABOLISM

SUMMARY TABLE: ANABOLISM VS CATABOLISM

	Anabolism	Catabolism
Direction	**Builds up**: small → large ("joining")	**Breaks down**: large → small
Energy (ATP)	**Requires** energy (ATP)	**Releases** energy (ATP); also heat and waste
Slide examples	**Protein synthesis** (amino acids → protein); **triacylglycerol** from glycerol + fatty acids; **photosynthesis**	**Carbohydrate digestion**: starch → polysaccharide → maltose and dextrin → glucose; **hydrolysis** of a polysaccharide to monosaccharides

- **Photosynthesis** (her anabolism example): plants are **producers**; light energy is required; $CO_2 + H_2O \rightarrow$ **glucose + O_2** (small → large).
- When organisms consume and catabolize the macromolecules, they are converted back to **CO_2 and H_2O**, releasing energy as **ATP**.
- **ATP ⇌ ADP:** when energy is used, ATP (adenosine **tri**phosphate) becomes **ADP (adenosine diphosphate)**; when energy is absorbed, ADP goes back to ATP.

Slide exercise (anabolic or catabolic)

(a) Synthesis of a protein from amino acids = **anabolic**. (b) Formation of a triacylglycerol from glycerol and fatty acids = **anabolic**. (c) Hydrolysis of a polysaccharide to monosaccharides = **catabolic** ("hydrolysis = break").

SITES OF METABOLISM

- **Anaerobic metabolism: no oxygen** involved; in the **cytoplasm**.
- **Aerobic metabolism: requires oxygen**; in the **mitochondria**. Oxygen is the **final acceptor of electrons** in food catabolism.

TYPES OF METABOLIC PATHWAYS

- **Linear:** a series of reactions that is **not repeated**. Example: **glycolysis** (lysis = breaking glucose); final product **pyruvate**; releases ATP.
- **Circular:** a **repeating** series in which the **final product is also an initial reactant**. Example: the **citric acid cycle (Krebs cycle)**: **oxaloacetate** reacts with **acetyl-CoA**, and oxaloacetate is formed again at the end.
- **Spiral:** a series of **repeated** reactions used to **break down (or build up)** a compound. Example: **β-oxidation of fatty acids**: an 8-carbon fatty acid loses 2 carbons per round (8 → 6 → 4 → 2) through **three** rounds, giving **4 acetyl-CoA, 3 $FADH_2$ and 3 NADH** ($FADH_2$ and NADH can be converted into ATP). The target is the **β-carbon** (next to the carbonyl).

THE GOAL OF METABOLISM

- The **ultimate goal**: convert the **chemical potential energy in food** into chemical potential energy in the form of **ATP**.

Slide exercise

The intermediate molecule responsible for producing energy in the body: **ATP** (not AMP, ADP or NADH).

WHAT TO REVIEW

- Bioinorganic vs bioorganic; the four biomolecules and their functions; CHON.
- The levels of organization in order, with the organelle and cell roles.
- Monomer ↔ polymer pairs; lipids are not polymers or macromolecules; informational macromolecules.
- The three bonds (peptide, glycosidic, phosphodiester), their directions (N→C, 1→4, 5'→3') and dehydration synthesis/condensation.
- Metabolism terms (pathway, metabolite, enzyme); anabolism vs catabolism with examples; ATP/ADP; aerobic vs anaerobic sites; linear, circular, spiral pathways with examples.